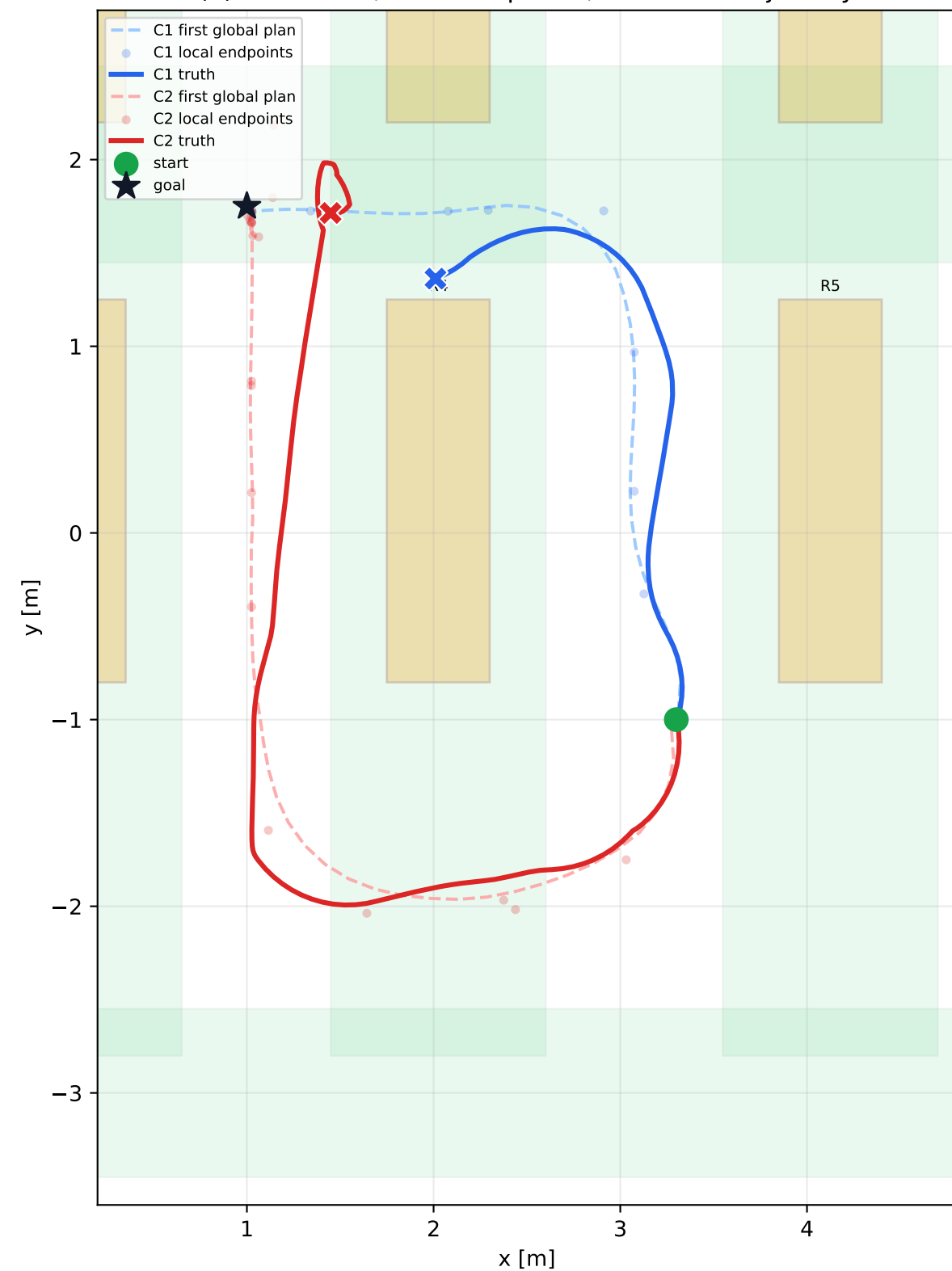


F45 - Tracking yaw + local solver timing diagnostic

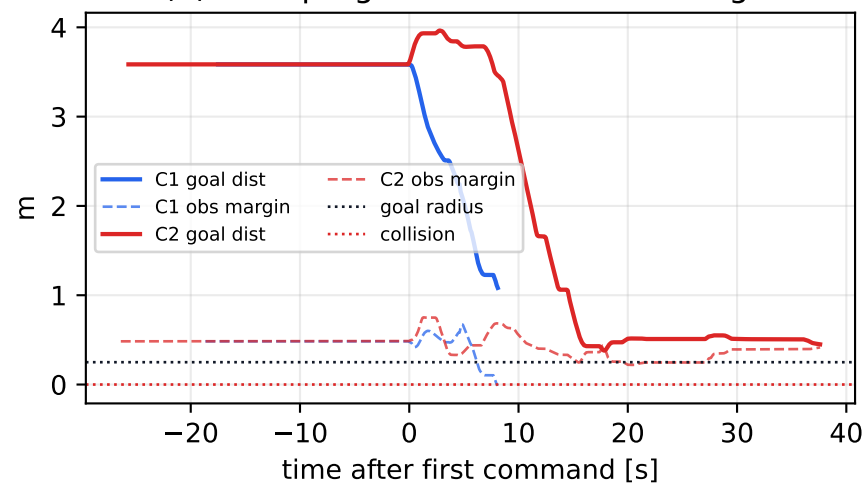
R4

R5

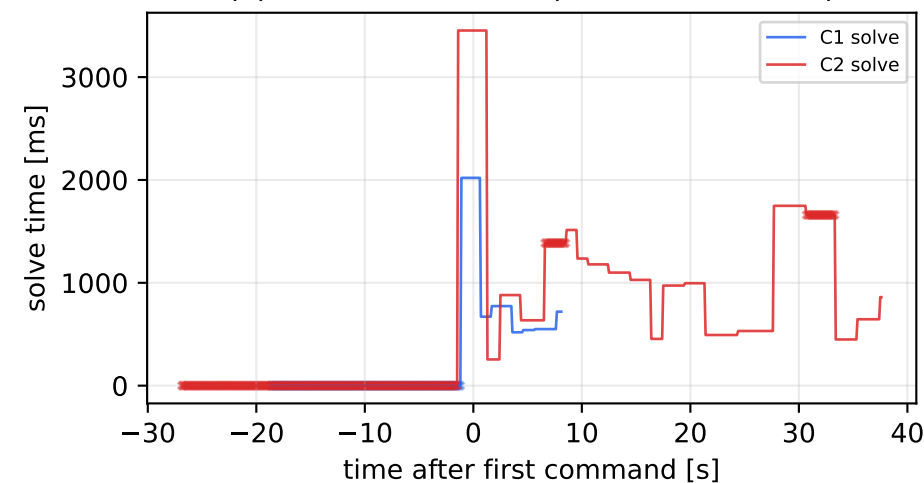
(a) First route, local endpoints, executed trajectory



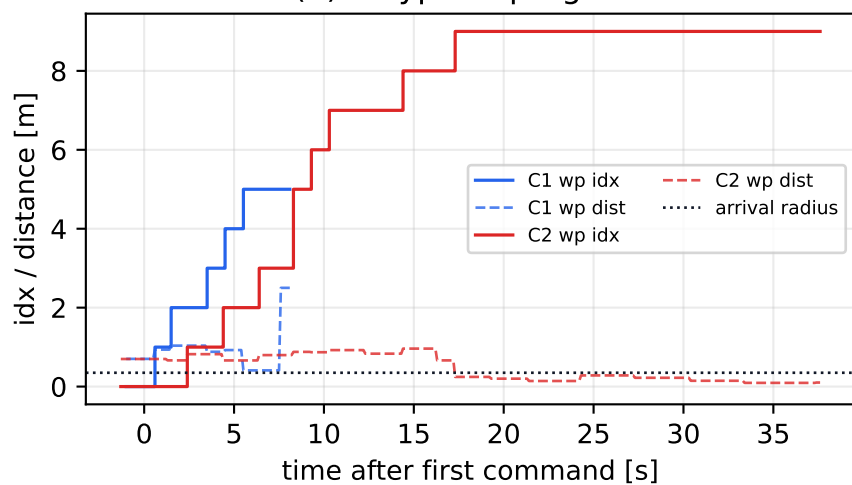
(b) Goal progress and obstacle margin



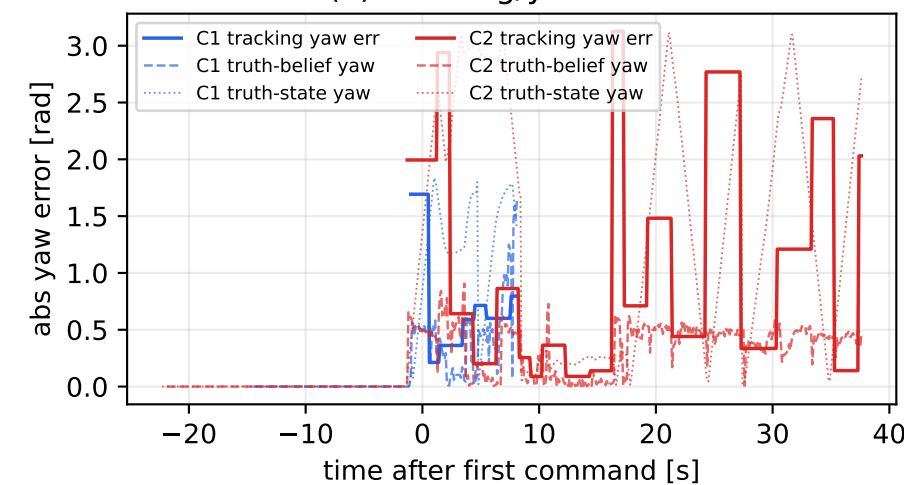
(c) Local solve time (x = non-success)



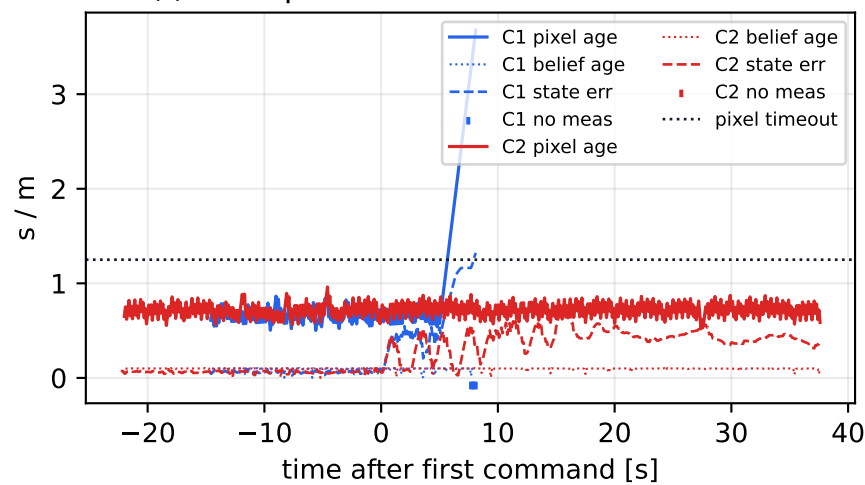
(d) Waypoint progress



(e) Heading/yaw errors



(f) Perception freshness and localization



F45 controlled tracking-yaw smoke

C1: collision, path=3.68m, min_goal=1.08m
err=0.262m, min obs=-0.013m
solve μ /p90=828/1272 ms, 19 ms/eval
maxiter/non-success rows=0%
yaw track μ /p90=0.72/1.69 rad
pixel age p90/max=1.41/3.68s

C2: stuck, path=7.04m, min_goal=0.38m
err=0.280m, min obs=0.220m
solve μ /p90=1074/1668 ms, 25 ms/eval
maxiter/non-success rows=3%
yaw track μ /p90=1.04/2.77 rad
pixel age p90/max=0.82/0.96s

Interpretation:

1. Spawn yaw is clean; yaw problems are runtime/tracking, not init.
2. Global plans still choose the expected route families.
3. Local H15/maxiter20 is too budget-starved: frequent maxiter hits.
4. C1 additionally suffers stale camera belief before collision.
5. C2 localizes better but still collides: local tracking/clearance remains failing.